SBT-DF204 Computer Forensics Case Studies

Case Study 2

Reconstructing Suspected Drug Trafficking From Chrome Web History

Individual Forensic Investigation Report

Student full name	Daniel Ukpong
Registration number	Fwsd2511186
Course code	SBT-DF204
Assessment	Case Study 2: Individual browser history investigation
Evidence file	History (Google Chrome History SQLite database)
Date of report	11 October 2026

This report was prepared for an authorised academic case study. The database is controlled training material. No live website, account, message, exchange, wallet or transaction referenced in the history was visited or contacted.

1. Case Reference and Student Identification

Item	Detail
Student full name	Daniel Ukpong
Registration number	Fwsd2511186
Course code	SBT-DF204 Computer Forensics Case Studies
Case title	Case Study 2: Reconstructing Suspected Drug Trafficking From Chrome Web History
Evidence file	History (Chrome History SQLite database), supplied in the study material section
Analysis platform	Kali Linux; sqlite3 (read-only mode) and DB Browser for SQLite; sha256sum
Report file name	SBTDF204_CaseStudy2_Fwsd2511186.docx

1.1 Scenario summary

Investigators received a Google Chrome History file from a device associated with a suspect. The case hypothesis is that the browser may have been used to advertise controlled drugs, communicate with an interested buyer, exchange payment information, verify a cryptocurrency transaction, and retain proof of payment. The task is to assess the history evidence and build a defensible, time-ordered account, while keeping apart what the browser history directly records and what is inferred about the user or the alleged activity.

1.2 Scope and authorisation

The analysis used only the supplied History database. It is a forensic interpretation of stored records. No live posting, person, email account, exchange, wallet, transaction identifier or external platform was visited, searched or contacted. Strings that look like session tokens in long URLs were not reproduced; URLs in this report are shortened for that reason.

1.3 Materials used

- The Chrome History SQLite database named History (Section 3).
- The Case Study 2 assessment brief, including its starting SQL and its transition interpretation notes.

2. Executive Conclusion

The History database records one continuous browsing session on 19 April 2022, from 13:54:18 to 15:02:56 UTC (1 hour 8 minutes 38 seconds), in which a single Chrome profile logged in to Craigslist and completed the posting workflow for a listing titled "cheaper than Rx supplements" in the health and beauty category (management page 7473121658, listing

opened at 13:58:31 UTC). The same profile then opened a Gmail inbox and an email thread titled "order" (14:04:12), later opened a different Gmail thread whose subject contains a Bitcoin transaction identifier beginning 517b2156 (14:56:47), followed an Imgur link from that thread, and saved an image named proof_of_payment.png to C:\Users\FSCS_User\Desktop (download record 1, 33,844 bytes, 14:56:58 to 14:57:10 UTC). Within the next six minutes the same identifier was looked up on blockchain.com (14:59:06) and, through a link opened from a Kraken page, on mempool.space (15:01:33).

This sequence is consistent with the case hypothesis of a posting, communication with a counterpart, an exchange of payment information, verification of a cryptocurrency transaction and retention of proof of payment. The history does not prove any of those things. It stores page titles, URLs, times and navigation links, not page content. It does not show what the listing said, what any email contained, what the image depicts, who sent or received any message, or that a transaction was completed. The listing title refers to supplements in a health and beauty category. The controlled-drug element comes from the case hypothesis, not from the record.

Confidence is moderate to high that this one Chrome profile performed the recorded sequence, low to moderate for any statement about the person who used it, and low for any statement that the activity was drug trafficking. The History file identifies a browser profile, not a human being. The findings are an investigative lead that needs corroboration from device, account and platform records.

3. Evidence Acquisition and Integrity

3.1 Evidence record

Item	Value
Original file name	History (Chrome browsing-history database, no file extension)
Additional copy supplied	Chrome_History_Cryptocurrency_Lab.db. Its SHA-256 value is identical to History, so it is a duplicate copy of the same evidence and not a separate exhibit.
Source	Case study material, study material section, as stated in the assessment brief
File size	196,608 bytes
Format	SQLite 3.x database, UTF-8 text, last written with SQLite 3.36.0
Date placed on analysis workstation	8 October 2026
Original location	~/SBT-DF204-CS2/original/History (permissions changed to read-only, mode 444)
Working copy	~/SBT-DF204-CS2/working/History_working, created with cp from the original
SHA-256 (original and working copy)	b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19

Item	Value
Tools	sha256sum; sqlite3 with the -readonly option; DB Browser for SQLite opened read-only
Output folder	~/SBT-DF204-CS2/output (hash records and CSV export of the visit timeline)

3.2 Hash comparison

Item	SHA-256	Match
original/History	b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19	Yes
working/History_working	b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19	Yes

The two values are identical, which shows that the working copy is bit-for-bit identical to the supplied file.

3.3 Handling statement

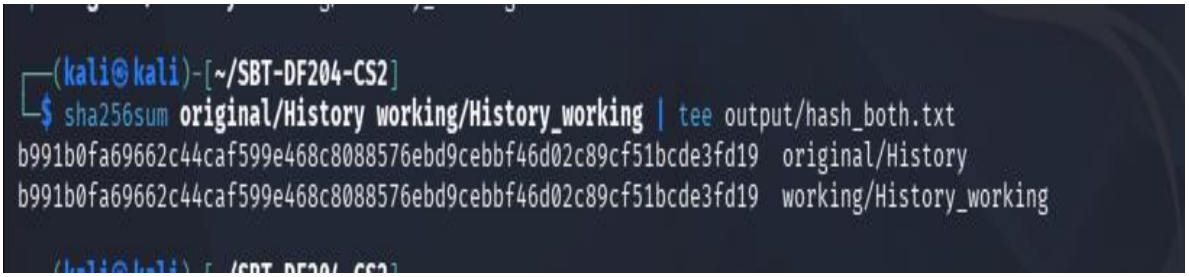
The supplied file was moved into an evidence folder, hashed, and made read-only. A working copy was made and every query was run on the working copy. The sqlite3 client was started with -readonly and DB Browser for SQLite was used to view the working copy. No statement that modifies data (INSERT, UPDATE, DELETE, VACUUM or Write Changes) was run. Any journal file that a viewer might create would exist only beside the working copy, never beside the original.

Figure S01. Hash of the original file and its size. [Evidence ID: E01]



Command Use: cd ~/SBT-DF204-CS2 ; ls -l original/History ; sha256sum original/History

Figure S02. Both hashes after the original was made read-only and the working copy was created. [Evidence ID: E01]



Command Use: sha256sum original/History working/History_working ; ls -l original working

4. Database Method

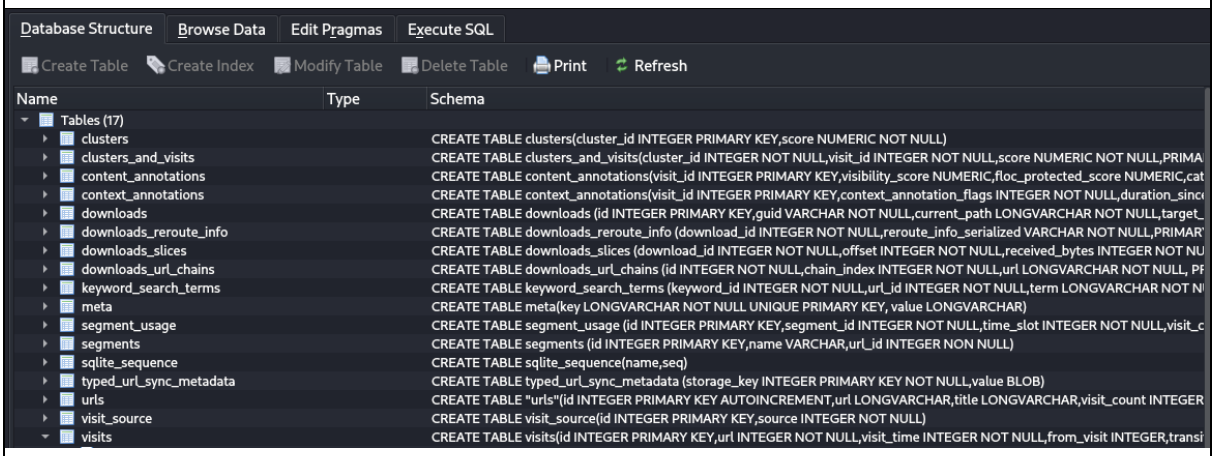
4.1 Tables and fields examined

The database contains 17 tables. The inspection used the structure view first, because Chrome versions differ, and then four tables that carry the evidence. The record counts are 57 URLs, 74 visits, 1 download and 1 download URL chain.

Table	Fields used	Role in this case
Urls	id, url, title, visit_count, typed_count, last_visit_time, hidden	One row for each distinct URL. Holds the title. The title is the most recent one Chrome stored for that URL .
Visits	id, url, visit_time, from_visit, transition, visit_duration, opener_visit	One row for each visit. url points to urls.id. from_visit points to the visit it followed. opener_visit points to the visit from which a tab was opened. transition describes how the browser arrived.
Downloads	id, target_path, current_path, start_time, end_time, received_bytes, total_bytes, state, mime_type, tab_url, referrer, last_modified, etag, opened, hash	Describes each saved file: where it was saved, size, state, times and the page it came from.
downloads_url_chains	id, chain_index, url	The actual address the file was downloaded from, in order, for each download id.

Table	Fields used	Role in this case
keyword_search_terms	keyword_id, url_id, term, normalized_term	Search terms Chrome extracted from the omnibox search pages. Used as a cross-check on the omnibox transitions.

Figure S03. Database Structure view listing the 17 tables. [Evidence ID: E02]



Command Use: sqlite3 -readonly working/History_working ".tables"

Figure S04. Field definitions of urls, visits, downloads and downloads_url_chains. [Evidence ID: E02]

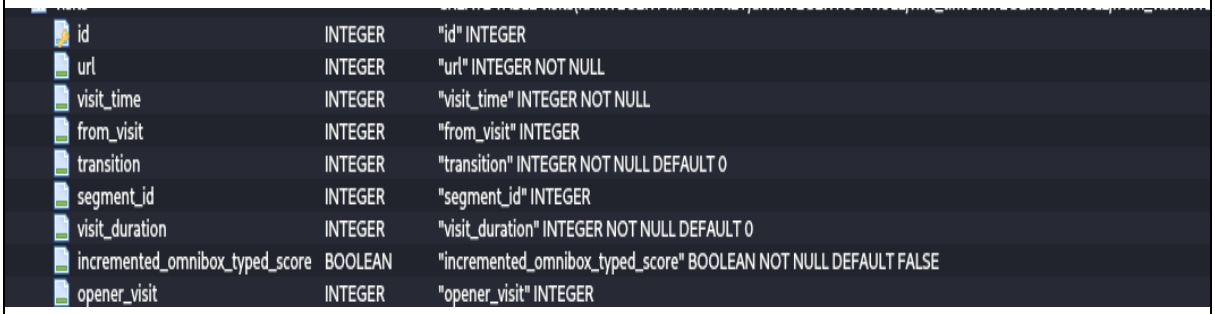


Figure S04b. Record counts of the four tables. [Evidence ID: E02]



Output: (expect urls 57, visits 74, downloads 1, downloads_url_chains 1)

4.2 Timestamp conversion

Chrome stores visit_time, start_time and end_time as WebKit timestamps: microseconds since 1 January 1601 00:00 UTC. The conversion divides by 1,000,000 and adds the offset between 1601 and 1970, which SQLite supplies as strftime('%s','1601-01-01') (minus 11,644,473,600 seconds). All times in this report are UTC. Local time is shown only where stated, and is Lagos time (WAT, UTC+1), obtained by adding one hour. visit_duration is stored in microseconds and is shown here in seconds.

```
datetime(visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch')   -- UTC
datetime(visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch', '+1 hour')   -- Lagos (WAT)
```

4.3 Joins

- visits.url = urls.id attaches the URL and title to each visit.
- visits.from_visit = visits.id (joined to itself) and then urls.id gives the page the browser came from.
- downloads.id = downloads_url_chains.id gives the source address of each download.
- LIKE on url and title, and instr and substr on the same fields, were used to find the transaction identifier wherever it appears.

4.4 Transition interpretation

The low byte of transition (transition AND 255) gives the base type. The assessment materials give 0 for a link, 5 for omnibox-generated activity and 7 for a form submission. The higher bits are qualifiers: 0x10000000 starts a navigation chain, 0x20000000 ends it, 0x40000000 is a client redirect, 0x80000000 is a server redirect, and 0x02000000 marks entry from the address bar. In DB Browser the top bit appears as a negative decimal number because Chrome stores it in a signed 32-bit field, so the value was also printed as an eight-

digit hexadecimal number. The transition value was not relied on alone. It was read together with from_visit, opener_visit, titles and URLs.

Hex	Decimal	Count	Reading	Visit IDs
30000005	805306373	1	Omnibox (base 5), single-visit chain	1
32000005	838860805	3	Omnibox (base 5), address-bar flag, single-visit chain	22, 51, 59
30000007	805306375	7	Form submission (base 7), single-visit chain	8, 14, 15, 16, 17, 18, 19
30000000	805306368	21	Link, single-visit chain	2, 20, 21, 23, 34, 35, 36, 47, 52, 53, 54, 57, 58, 60 to 66, 71
10000000	268435456	12	Link, chain start	3, 6, 9, 11, 24, 31, 37, 43, 48, 49, 55, 67
80000000	-2147483648	13	Link, server redirect inside a chain	4, 12, 25 to 29, 32, 38 to 41, 69
A0000000	-1610612736	9	Link, server redirect, chain end	5, 7, 10, 13, 30, 33, 50, 56, 70
40000000	1073741824	5	Link, client redirect inside a chain	44, 45, 68, 72, 73
60000000	1610612736	3	Link, client redirect, chain end	42, 46, 74

5. Findings

5.1 Question 1: What evidence file did you acquire and how did you preserve it?

Finding. The file is History, 196,608 bytes, a SQLite database from the study material section. Its SHA-256 value is b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19. The original was made read-only (mode 444) and a working copy with the identical hash was used for every query. A second file named Chrome_History_Cryptocurrency_Lab.db has the same SHA-256 value and is a duplicate.

Evidence references. E01; Figures S01 and S02; Sections 3.1 to 3.3.

5.2 Question 2: What browser-history tables and fields are relevant?

Finding. The relevant tables are urls, visits, downloads and downloads_url_chains, with keyword_search_terms used as a cross-check. Their fields and roles are set out in Section 4.1. The counts returned by the working copy are shown below.

Table	Rows
Urls	57
Visits	74
Downloads	1
downloads_url_chains	1

Evidence references. E02; Figures S03, S04 and S04b; query

5.3 Question 3: When did the recorded web activity occur?

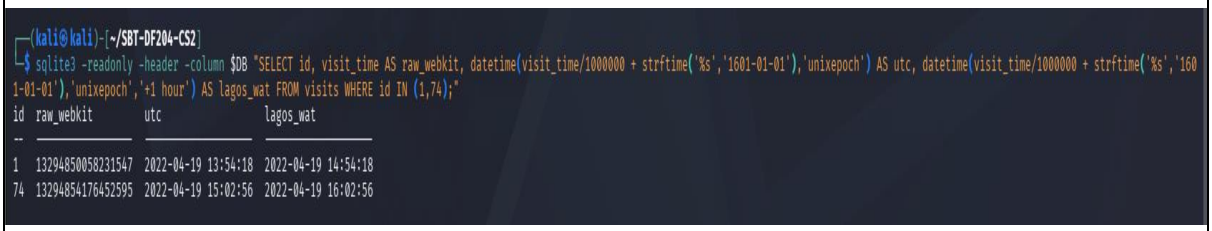
Finding. All 74 visits fall on 19 April 2022 between 13:54:18 UTC and 15:02:56 UTC, a span of 1 hour 8 minutes 38 seconds. In Lagos time (UTC+1) the same span is 14:54:18 to 16:02:56. The conversion is shown for the first and last visits.

Visit	Raw WebKit value	UTC	Lagos (UTC+1)	URL ID
1	13294850058231547	2022-04-19 13:54:18	2022-04-19 14:54:18	1
74	13294854176452595	2022-04-19 15:02:56	2022-04-19 16:02:56	37

The main time features are: a login and posting phase from 13:54:18 to 13:58:31; a Gmail phase from 13:58:38 to 14:05:59; no records for 50 minutes 48 seconds (14:05:59 to 14:56:47); then a payment-verification phase from 14:56:47 to 15:02:56. Section 6 gives the complete sequence.

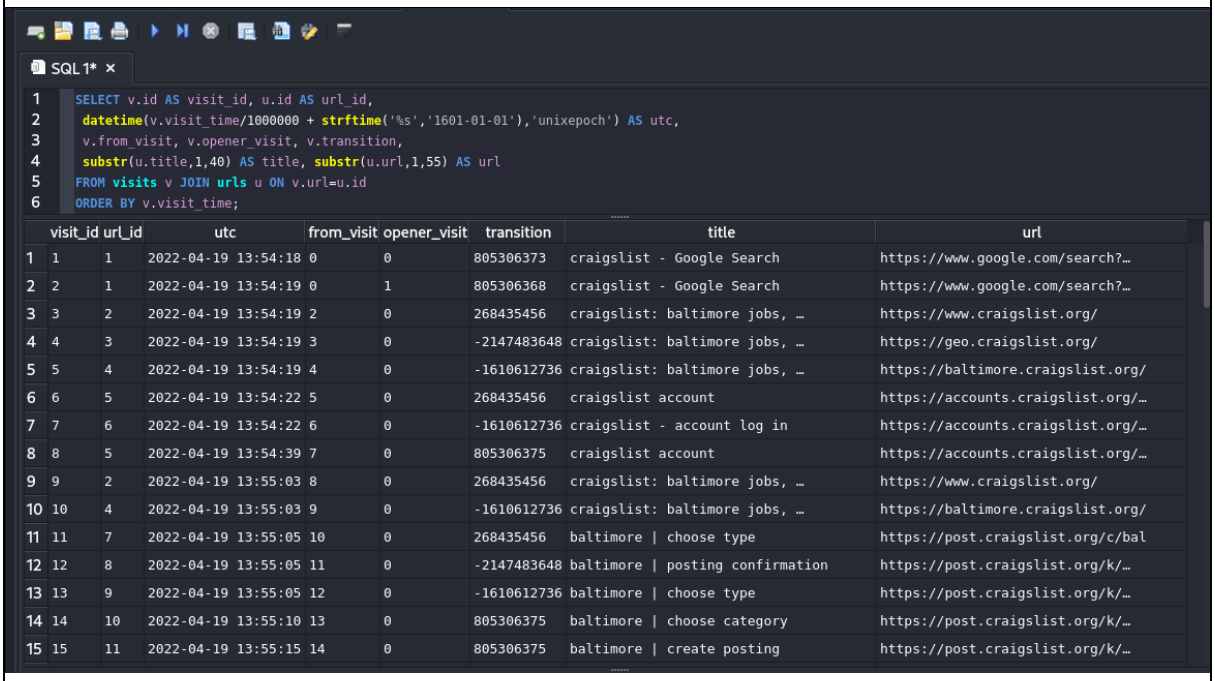
Evidence references. E03; Figures S05 and S06a to S06e; The complete visit-level timeline is in Appendix B.

Figure S05. Timestamp conversion of the first and last visits, in UTC and Lagos time. [Evidence ID: E03]



To capture: Run Q04 in Appendix D (expect visit 1 = 13294850058231547 = 13:54:18 UTC, visit 74 = 15:02:56 UTC).

Figure S06a. Visit timeline, visits 1 to 15. The SQL pane must be visible in this capture. [Evidence ID: E03]



SQL 1*

```
SELECT v.id AS visit_id, u.id AS url_id,
 datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS utc,
 v.from_visit, v.opener_visit, v.transition,
 substr(u.title,1,40) AS title, substr(u.url,1,55) AS url
FROM visits v JOIN urls u ON v.url=u.id
ORDER BY v.visit_time;
```

	visit_id	url_id	utc	from_visit	opener_visit	transition	title	url
1	1	1	2022-04-19 13:54:18	0	0	805306373	craigslist - Google Search	https://www.google.com/search?…
2	2	1	2022-04-19 13:54:19	0	1	805306368	craigslist - Google Search	https://www.google.com/search?…
3	3	2	2022-04-19 13:54:19	2	0	268435456	craigslist: baltimore jobs, …	https://www.craigslist.org/
4	4	3	2022-04-19 13:54:19	3	0	-2147483648	craigslist: baltimore jobs, …	https://geo.craigslist.org/
5	5	4	2022-04-19 13:54:19	4	0	-1610612736	craigslist: baltimore jobs, …	https://baltimore.craigslist.org/
6	6	5	2022-04-19 13:54:22	5	0	268435456	craigslist account	https://accounts.craigslist.org/…
7	7	6	2022-04-19 13:54:22	6	0	-1610612736	craigslist - account log in	https://accounts.craigslist.org/…
8	8	5	2022-04-19 13:54:39	7	0	805306375	craigslist account	https://accounts.craigslist.org/…
9	9	2	2022-04-19 13:55:03	8	0	268435456	craigslist: baltimore jobs, …	https://www.craigslist.org/
10	10	4	2022-04-19 13:55:03	9	0	-1610612736	craigslist: baltimore jobs, …	https://baltimore.craigslist.org/
11	11	7	2022-04-19 13:55:05	10	0	268435456	baltimore | choose type	https://post.craigslist.org/c/bal
12	12	8	2022-04-19 13:55:05	11	0	-2147483648	baltimore | posting confirmation	https://post.craigslist.org/k/…
13	13	9	2022-04-19 13:55:05	12	0	-1610612736	baltimore | choose type	https://post.craigslist.org/k/…
14	14	10	2022-04-19 13:55:10	13	0	805306375	baltimore | choose category	https://post.craigslist.org/k/…
15	15	11	2022-04-19 13:55:15	14	0	805306375	baltimore | create posting	https://post.craigslist.org/k/…

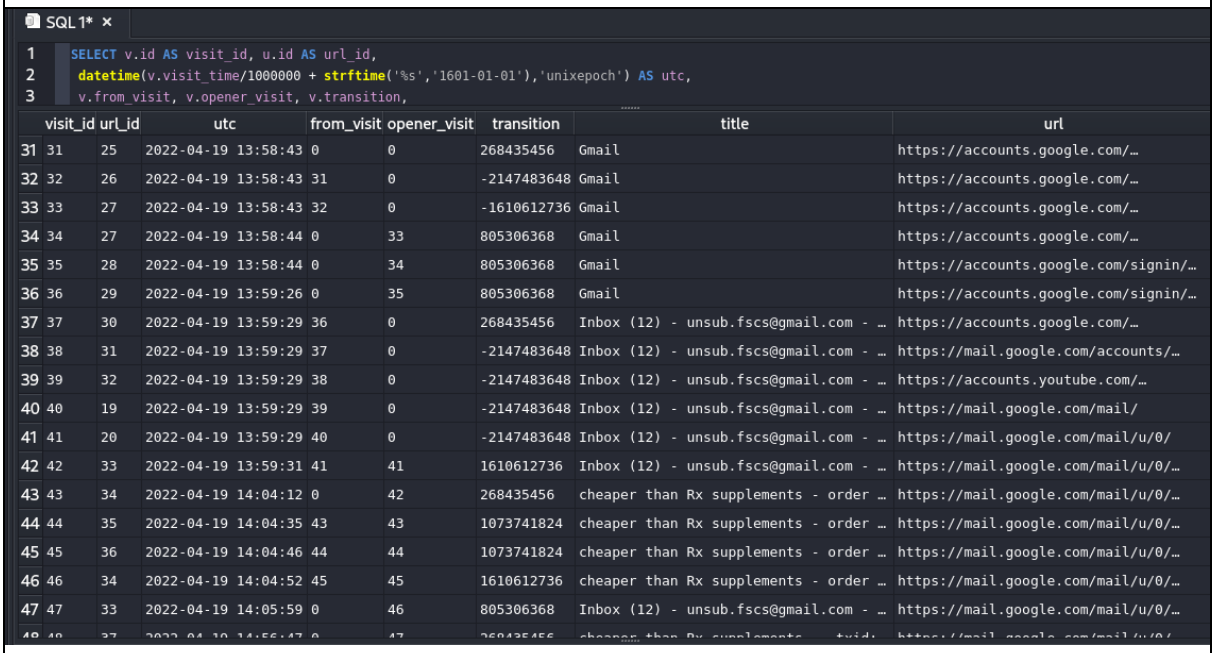
To screenshot: The query shows 74 rows. Capture visits 1 to 15.

Figure S06b. Visit timeline, visits 16 to 30 (same query, scrolled). [Evidence ID: E03]

	visit_id	url_id	utc	from_visit	opener_visit	transition	title	url
16	16	12	2022-04-19 13:58:13	15	0	805306375	baltimore | add map	https://post.craigslist.org/k/…
17	17	13	2022-04-19 13:58:17	16	0	805306375	baltimore | create posting	https://post.craigslist.org/k/…
18	18	14	2022-04-19 13:58:19	17	0	805306375	baltimore | create posting	https://post.craigslist.org/k/…
19	19	8	2022-04-19 13:58:22	18	0	805306375	baltimore | posting confirmation	https://post.craigslist.org/k/…
20	20	15	2022-04-19 13:58:27	19	0	805306368	baltimore | manage posting	https://post.craigslist.org/manage/…
21	21	16	2022-04-19 13:58:31	0	20	805306368	cheaper than Rx supplements - health…	https://baltimore.craigslist.org/hab…
22	22	17	2022-04-19 13:58:38	0	0	838860805	gmail - Google Search	https://www.google.com/search?…
23	23	17	2022-04-19 13:58:39	0	22	805306368	gmail - Google Search	https://www.google.com/search?…
24	24	18	2022-04-19 13:58:40	23	0	268435456	Gmail: Free, Private & Secure Email …	https://www.google.com/gmail/
25	25	19	2022-04-19 13:58:40	24	0	-2147483648	Inbox (12) - unsub.fscs@gmail.com - …	https://mail.google.com/mail/
26	26	20	2022-04-19 13:58:40	25	0	-2147483648	Inbox (12) - unsub.fscs@gmail.com - …	https://mail.google.com/mail/u/0/
27	27	21	2022-04-19 13:58:40	26	0	-2147483648	Gmail: Free, Private & Secure Email …	https://accounts.google.com/…
28	28	22	2022-04-19 13:58:40	27	0	-2147483648	Gmail: Free, Private & Secure Email …	https://mail.google.com/intl/en/mail…
29	29	23	2022-04-19 13:58:40	28	0	-2147483648	Gmail: Free, Private & Secure Email …	https://www.google.com/intl/en/mail/…
30	30	24	2022-04-19 13:58:40	29	0	-1610612736	Gmail: Free, Private & Secure Email …	https://www.google.com/gmail/about/

Screenshot: Same query Q05; scroll the grid.

Figure S06c. Visit timeline, visits 31 to 47 (same query, scrolled). [Evidence ID: E03]



SQL 1* ×

```
SELECT v.id AS visit_id, u.id AS url_id,
 datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS utc,
 v.from_visit, v.opener_visit, v.transition,
```

	visit_id	url_id	utc	from_visit	opener_visit	transition	title	url
31	31	25	2022-04-19 13:58:43	0	0	268435456	Gmail	https://accounts.google.com/…
32	32	26	2022-04-19 13:58:43	31	0	-2147483648	Gmail	https://accounts.google.com/…
33	33	27	2022-04-19 13:58:43	32	0	-1610612736	Gmail	https://accounts.google.com/…
34	34	27	2022-04-19 13:58:44	0	33	805306368	Gmail	https://accounts.google.com/…
35	35	28	2022-04-19 13:58:44	0	34	805306368	Gmail	https://accounts.google.com/signin/…
36	36	29	2022-04-19 13:59:26	0	35	805306368	Gmail	https://accounts.google.com/signin/…
37	37	30	2022-04-19 13:59:29	36	0	268435456	Inbox (12) - unsub.fscs@gmail.com - …	https://accounts.google.com/…
38	38	31	2022-04-19 13:59:29	37	0	-2147483648	Inbox (12) - unsub.fscs@gmail.com - …	https://mail.google.com/accounts/…
39	39	32	2022-04-19 13:59:29	38	0	-2147483648	Inbox (12) - unsub.fscs@gmail.com - …	https://accounts.youtube.com/…
40	40	19	2022-04-19 13:59:29	39	0	-2147483648	Inbox (12) - unsub.fscs@gmail.com - …	https://mail.google.com/mail/
41	41	20	2022-04-19 13:59:29	40	0	-2147483648	Inbox (12) - unsub.fscs@gmail.com - …	https://mail.google.com/mail/u/0/
42	42	33	2022-04-19 13:59:31	41	41	1610612736	Inbox (12) - unsub.fscs@gmail.com - …	https://mail.google.com/mail/u/0/…
43	43	34	2022-04-19 14:04:12	0	42	268435456	cheaper than Rx supplements - order …	https://mail.google.com/mail/u/0/…
44	44	35	2022-04-19 14:04:35	43	43	1073741824	cheaper than Rx supplements - order …	https://mail.google.com/mail/u/0/…
45	45	36	2022-04-19 14:04:46	44	44	1073741824	cheaper than Rx supplements - order …	https://mail.google.com/mail/u/0/…
46	46	34	2022-04-19 14:04:52	45	45	1610612736	cheaper than Rx supplements - order …	https://mail.google.com/mail/u/0/…
47	47	33	2022-04-19 14:05:59	0	46	805306368	Inbox (12) - unsub.fscs@gmail.com - …	https://mail.google.com/mail/u/0/…

To capture: Same query Q05; scroll the grid.

5.4 Question 4: How did the browser reach the material URLs?

Base types. Of the 74 visits, 63 are links (base 0), 4 are omnibox-generated (base 5) and 7 are form submissions (base 7). No visit has base 1 (a typed address), and typed_count is 0 for every URL.

Base type	Visits and reading
5, omnibox	Visits 1 (craigslist), 22 (gmail), 51 (blockchain explorer) and 59 (kraken). Each is a search typed in the address bar. Visits 22, 51 and 59 carry the address-bar flag (hex 32000005); visit 1 is hex 30000005 without it. The keyword_search_terms table lists the same four terms (craigslist, gmail, blockchain explorer, kraken).
7, form submit	Visit 8 (Craigslist account page after login) and visits 14 to 19, the six steps of the posting wizard: category, create posting, add map, image step, preview, and the confirmation at visit 19.
0, link	All other visits, including every redirect inside a chain and every page opened by clicking within a site.

Redirect and navigation chains. The chain qualifiers show which visits are automatic redirects and which are page loads. Counting every redirect step as user action would overstate activity; for example visits 24 to 30 all occurred within the same second.

Visits	Qualifiers	Pages reached
3 to 5	start, server, end	craigslist.org, geo.craigslist.org, baltimore.craigslist.org
6 to 7	start, end	accounts.craigslist.org/login/home, then the Craigslist login page

Visits	Qualifiers	Pages reached
9 to 10	start, end	craigslist.org, then baltimore.craigslist.org
11 to 13	start, server, end	post.craigslist.org/c/bal, a posting-confirmation address, then the posting step choose type
24 to 30	start, 5 server, end	google.com/gmail/, mail.google.com/mail/, mail/u/0/, the Google sign-in service, Gmail help pages, ending at google.com/gmail/about/
31 to 33	start, server, end	Google account chooser and sign-in service pages
37 to 42	start, 4 server, client end	accounts.google.com sign-in completion, sign-in cookie hand-off pages, then the Gmail inbox
43 to 46	start, 2 client, client end	Email thread "order", two compose addresses (?compose=), return to the thread
49 to 50	start, server end	Google link wrapper (source=gmail), then imgur.com/dTgrkP7
55 to 56	start, server end	blockchain.com search for the identifier, then the Bitcoin transaction page
67 to 70	start, client, server, end	Kraken redirect link, then mempool.space/tx/ for the same identifier
48, 72 to 74	start; then 2 client, client end	Visit 72 follows visit 48 (from_visit 48), so the compose redirects at 15:02 continue from the thread opened at 14:56:47

Opener links. Twenty-eight visits have a non-zero opener_visit, which records the visit from which a page or tab was opened. Three are important to the case: the live listing (visit 21) was opened from the management page (visit 20); the Imgur page (visit 49) was opened from the email thread (visit 48); and the mempool link (visit 67) was opened from the Kraken funding page (visit 66). Separately, 53 of the 74 visits have a non-zero from_visit, which links them to the visit they followed (the other 21 start a new navigation).

Limit. A transition value describes how Chrome arrived at a page. It does not show who clicked, and a compose or redirect address does not show that a message was written or sent.

Evidence references. E04; Figures S07, S08a, S08b, S08c and S08d; queries Q06 to Q09 and Q19.

Figure S07. Transition decode: base type, qualifier flags and hexadecimal value for every visit (three captures if needed). [Evidence ID: E04]

SQL 1* ×

```
4   CASE v.transition & 255 WHEN 0 THEN 'LINK' WHEN 5 THEN 'OMNIBOX' WHEN 7 THEN 'FORM_SUBMIT' ELSE 'other' END AS base_name,
5   (v.transition & 268435456)<>0 AS chain_start,
6   (v.transition & 536870912)<>0 AS chain_end,
7   (v.transition & 1073741824)<>0 AS client_redir,
8   (v.transition & 2147483648)<>0 AS server_redir,
9   (v.transition & 33554432)<>0 AS addr_bar
10  FROM visits v ORDER BY v.visit_time;
```

	id	from_visit	opener_visit	tr_hex	base	base_name	chain_start	chain_end	client_redir	server_redir	addr_bar
1	1	0	0	30000005	5	OMNIBOX	1	1	0	0	0
2	2	0	1	30000000	0	LINK	1	1	0	0	0
3	3	2	0	10000000	0	LINK	1	0	0	0	0
4	4	3	0	80000000	0	LINK	0	0	0	1	0
5	5	4	0	A0000000	0	LINK	0	1	0	1	0
6	6	5	0	10000000	0	LINK	1	0	0	0	0
7	7	6	0	A0000000	0	LINK	0	1	0	1	0
8	8	7	0	30000007	7	FORM_SUBMIT	1	1	0	0	0
9	9	8	0	10000000	0	LINK	1	0	0	0	0
10	10	9	0	A0000000	0	LINK	0	1	0	1	0
11	11	10	0	10000000	0	LINK	1	0	0	0	0
12	12	11	0	80000000	0	LINK	0	0	0	1	0
13	13	12	0	A0000000	0	LINK	0	1	0	1	0
14	14	13	0	30000007	7	FORM_SUBMIT	1	1	0	0	0
15	15	14	0	30000007	7	FORM_SUBMIT	1	1	0	0	0

Screenshot: Capture the 74 rows in three scrolled images and keep the SQL visible in the first.

Figure S08a. Distribution of base transition types (0 = 63, 5 = 4, 7 = 7). [Evidence ID: E04]

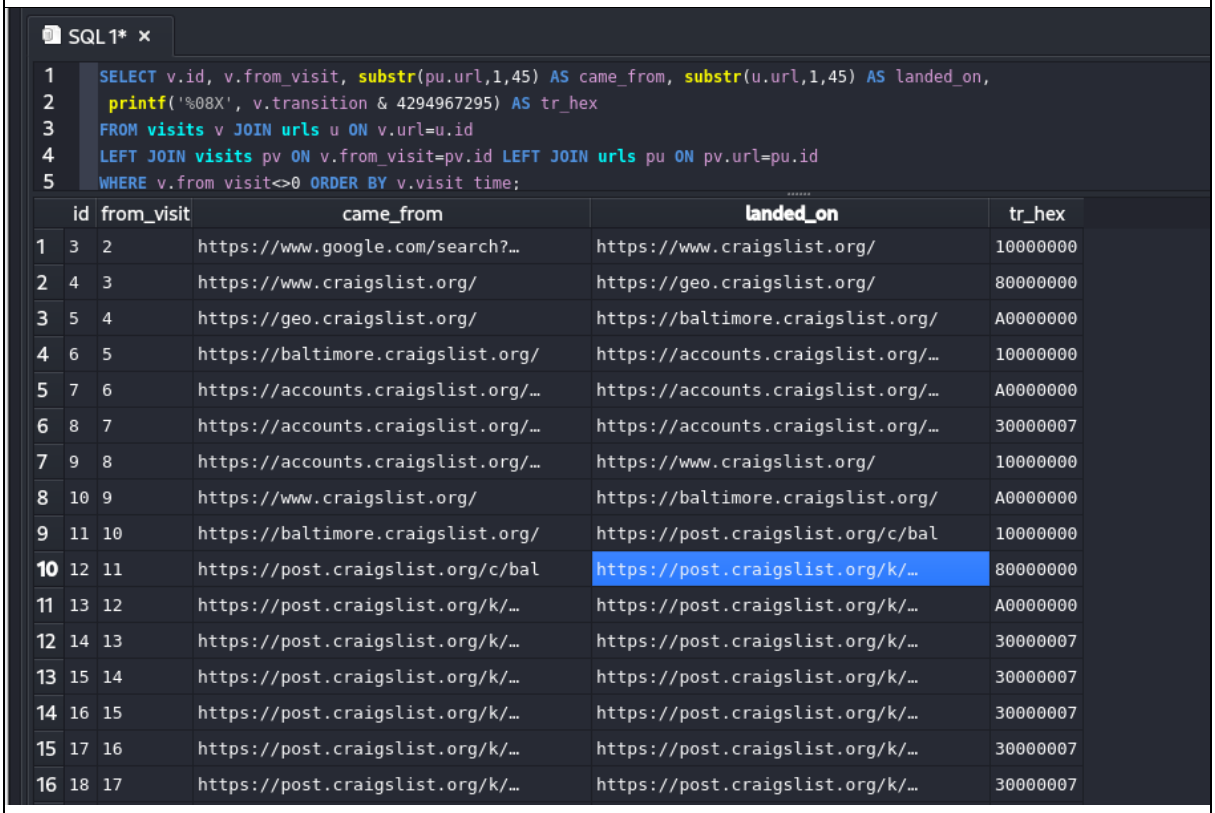
Database Structure | Browse Data | Edit Pragmas | Execute SQL

SQL 1* ×

```
1   SELECT transition & 255 AS base, COUNT(*) AS n FROM visits GROUP BY base;
```

	base	n
1	0	63
2	5	4
3	7	7

Figure S08b. from_visit chain with the page the browser came from and the page it landed on (53 rows). [Evidence ID: E04]



SQL 1* ×

```
SELECT v.id, v.from_visit, substr(pu.url,1,45) AS came_from, substr(u.url,1,45) AS landed_on,
 printf('%08X', v.transition & 4294967295) AS tr_hex
FROM visits v JOIN urls u ON v.url=u.id
LEFT JOIN visits pv ON v.from_visit=pv.id LEFT JOIN urls pu ON pv.url=pu.id
WHERE v.from_visit<>0 ORDER BY v.visit_time;
```

	id	from_visit	came_from	landed_on	tr_hex
1	3	2	https://www.google.com/search?…	https://www.craigslist.org/	10000000
2	4	3	https://www.craigslist.org/	https://geo.craigslist.org/	80000000
3	5	4	https://geo.craigslist.org/	https://baltimore.craigslist.org/	A0000000
4	6	5	https://baltimore.craigslist.org/	https://accounts.craigslist.org/…	10000000
5	7	6	https://accounts.craigslist.org/…	https://accounts.craigslist.org/…	A0000000
6	8	7	https://accounts.craigslist.org/…	https://accounts.craigslist.org/…	30000007
7	9	8	https://accounts.craigslist.org/…	https://www.craigslist.org/	10000000
8	10	9	https://www.craigslist.org/	https://baltimore.craigslist.org/	A0000000
9	11	10	https://baltimore.craigslist.org/	https://post.craigslist.org/c/bal	10000000
10	12	11	https://post.craigslist.org/c/bal	https://post.craigslist.org/k/…	80000000
11	13	12	https://post.craigslist.org/k/…	https://post.craigslist.org/k/…	A0000000
12	14	13	https://post.craigslist.org/k/…	https://post.craigslist.org/k/…	30000007
13	15	14	https://post.craigslist.org/k/…	https://post.craigslist.org/k/…	30000007
14	16	15	https://post.craigslist.org/k/…	https://post.craigslist.org/k/…	30000007
15	17	16	https://post.craigslist.org/k/…	https://post.craigslist.org/k/…	30000007
16	18	17	https://post.craigslist.org/k/…	https://post.craigslist.org/k/…	30000007

Captured in scrolled images.

Figure S08c. Visits opened from another visit (opener_visit, 28 rows). [Evidence ID: E04]

```
SELECT v.id, v.opener_visit, substr(u.url,1,50) AS landed_on
FROM visits v JOIN urls u ON v.url=u.id WHERE v.opener_visit<>0 ORDER BY v.visit_time;
```

	id	opener_visit	landed_on
1	2	1	https://www.google.com/search?…
2	21	20	https://baltimore.craigslist.org/hab…
3	23	22	https://www.google.com/search?…
4	34	33	https://accounts.google.com/…
5	35	34	https://accounts.google.com/signin/…
6	36	35	https://accounts.google.com/signin/…
7	42	41	https://mail.google.com/mail/u/0/…
8	43	42	https://mail.google.com/mail/u/0/…
9	44	43	https://mail.google.com/mail/u/0/…
10	45	44	https://mail.google.com/mail/u/0/…
11	46	45	https://mail.google.com/mail/u/0/…
12	47	46	https://mail.google.com/mail/u/0/…
13	48	47	https://mail.google.com/mail/u/0/…
14	49	48	https://www.google.com/url?q=https:/…
15	52	51	https://www.google.com/search?…
16	54	53	https://www.blockchain.com/explorer
17	57	56	https://www.blockchain.com/btc/tx/…

5.5 Question 5: What evidence relates to the suspected sale or posting?

Finding. Visits 1 to 21 are the Craigslist activity (21 rows are returned because the Google search addresses at visits 1 and 2 also contain the word craigslist). The records show an omnibox search, an account login, a posting workflow of form submissions, a management page and the live listing.

Visit	URL ID	UTC	Trans.	From	Open	Title	URL
1	1	13:54:18	30000005	0	0	craigslist - Google Search	https://www.google.com/search?q=craigslist&rlz=1C...
2	1	13:54:19	30000000	0	1	craigslist - Google Search	https://www.google.com/search?q=craigslist&rlz=1C...
3	2	13:54:19	10000000	2	0	craigslist: baltimore jobs, apartme...	https://www.craigslist.org/

Visit	URL ID	UTC	Trans.	From	Open	Title	URL
4	3	13:54:19	80000000	3	0	craigslist: baltimore jobs, apartme...	https://geo.craigslist.org/
5	4	13:54:19	A0000000	4	0	craigslist: baltimore jobs, apartme...	https://baltimore.craigslist.org/
6	5	13:54:22	10000000	5	0	craigslist account	https://accounts.craigslist.org/login/home
7	6	13:54:22	A0000000	6	0	craigslist - account log in	https://accounts.craigslist.org/login?rp=%2Flogin...
8	5	13:54:39	30000007	7	0	craigslist account	https://accounts.craigslist.org/login/home
9	2	13:55:03	10000000	8	0	craigslist: baltimore jobs, apartme...	https://www.craigslist.org/
10	4	13:55:03	A0000000	9	0	craigslist: baltimore jobs, apartme...	https://baltimore.craigslist.org/
11	7	13:55:05	10000000	10	0	baltimore | choose type	https://post.craigslist.org/c/bal
12	8	13:55:05	80000000	11	0	baltimore | posting confirmation	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbe...
13	9	13:55:05	A0000000	12	0	baltimore | choose type	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbe...
14	10	13:55:10	30000007	13	0	baltimore | choose category	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbe...
15	11	13:55:15	30000007	14	0	baltimore | create posting	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbe...
16	12	13:58:13	30000007	15	0	baltimore | add map	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbe...
17	13	13:58:17	30000007	16	0	baltimore | create posting	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbe...
18	14	13:58:19	30000007	17	0	baltimore | create posting	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbe...
19	8	13:58:22	30000007	18	0	baltimore | posting confirmation	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbe...

Visit	URL ID	UTC	Trans.	From	Open	Title	URL
20	15	13:58:27	30000000	19	0	baltimore | manage posting	https://post.craigslist.org/manage/7473121658
21	16	13:58:31	30000000	0	20	cheaper than Rx supplements - healt...	https://baltimore.craigslist.org/hab/d/baltimore-...

The records support the following observations:

- **Login.** Visits 6 to 8, 13:54:22 to 13:54:39. The login page (visit 7) was open for 17.5 seconds and visit 8, a form submission, followed at 13:54:39.
- **Posting workflow.** Visits 11 to 19, 13:55:05 to 13:58:22, on post.craigslist.org. Visits 14 to 19 are form submissions (hex 30000007): choose category, create posting, add map, image step, preview and confirmation.
- **Time spent writing.** The create posting page (visit 15) stayed open for 177.7 seconds (about 2 minutes 58 seconds), from 13:55:15 until the next step at 13:58:13. This is consistent with a form being completed, but the history does not store what was typed.
- **Confirmation.** The confirmation of the posting is visit 19 (13:58:22), a form submission to URL ID 8. Visit 12 carries the same title, "posting confirmation", but it is an intermediate server redirect inside the chain 11 to 13 at 13:55:05, at the start of the wizard. It should not be read as the moment the advert was published.
- **Management page and live listing.** Visit 20 (13:58:27) is "manage posting" for posting 7473121658. Visit 21 (13:58:31) is the listing page, "cheaper than Rx supplements - health and beauty - by owner", opened from visit 20 (opener_visit 20).

What the record supports. A listing was created and published under a Craigslist account in the Baltimore area, and the profile viewed it four seconds after opening the management page.

What the record does not show. The wording and any images of the advert, the price, the contact details, and the reason the listing refers to Rx supplements. The title and category do not themselves show an advert for controlled drugs. That interpretation comes from the case hypothesis. The record also does not identify the account holder.

Evidence references. E05; Figure S09; query Q10.

Figure S09. Craigslist visits: login, posting workflow, management page and listing (21 rows). [Evidence ID: E05]

```
SELECT v.id, u.id AS url_id,
 datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS utc,
 v.transition & 255 AS base, u.title, substr(u.url,1,70) AS url
FROM visits v JOIN urls u ON v.url=u.id
WHERE u.url LIKE '%craigslist%' ORDER BY v.visit_time;
```

	id	url_id	utc	base	title	url
1	1	1	2022-04-19 13:54:18	5	craigslist - Google Search	https://www.google.com/search?…
2	2	1	2022-04-19 13:54:19	0	craigslist - Google Search	https://www.google.com/search?…
3	3	2	2022-04-19 13:54:19	0	craigslist: baltimore jobs, …	https://www.craigslist.org/
4	4	3	2022-04-19 13:54:19	0	craigslist: baltimore jobs, …	https://geo.craigslist.org/
5	5	4	2022-04-19 13:54:19	0	craigslist: baltimore jobs, …	https://baltimore.craigslist.org/
6	6	5	2022-04-19 13:54:22	0	craigslist account	https://accounts.craigslist.org/…
7	7	6	2022-04-19 13:54:22	0	craigslist - account log in	https://accounts.craigslist.org/…
8	8	5	2022-04-19 13:54:39	7	craigslist account	https://accounts.craigslist.org/…
9	9	2	2022-04-19 13:55:03	0	craigslist: baltimore jobs, …	https://www.craigslist.org/
10	10	4	2022-04-19 13:55:03	0	craigslist: baltimore jobs, …	https://baltimore.craigslist.org/
11	11	7	2022-04-19 13:55:05	0	baltimore | choose type	https://post.craigslist.org/c/bal
12	12	8	2022-04-19 13:55:05	0	baltimore | posting confirmation	https://post.craigslist.org/k/…
13	13	9	2022-04-19 13:55:05	0	baltimore | choose type	https://post.craigslist.org/k/…
14	14	10	2022-04-19 13:55:10	7	baltimore | choose category	https://post.craigslist.org/k/…
15	15	11	2022-04-19 13:55:15	7	baltimore | create posting	https://post.craigslist.org/k/…
16	16	12	2022-04-19 13:58:13	7	baltimore | add map	https://post.craigslist.org/k/…

5.6 Question 6: What evidence relates to communications and payment activity?

5.6.1 Email access and the order thread

Visit 22 (13:58:38) is an omnibox search for gmail, seven seconds after the listing was opened. Visits 24 to 30 are a redirect chain that ends at the public Gmail about page, followed by the account chooser and sign-in pages (visits 31 to 36), a password-challenge page at 13:59:26 (visit 36) and the completion of sign-in at 13:59:29 (visit 37). Visits 38 to 42 follow the hand-off pages and reach the inbox at 13:59:31 (visit 42). The Gmail account label in the page titles is unsub.fscs@gmail.com.

The page titles for visits 25 and 26 show "Inbox (12)", but Chrome keeps only the latest title for each URL, and the same URLs were loaded again at visits 40 and 41. The earlier visits therefore must not be read as showing the inbox at 13:58:40.

Visit	URL ID	UTC	Trans.	From	Open	Title	URL
22	17	13:58:38	32000005	0	0	gmail - Google Search	https://www.google.com/search?q=gmail&rlz=1C1ONGR...
30	24	13:58:40	A0000000	29	0	Gmail: Free, Private & Secure Email...	https://www.google.com/gmail/about/
36	29	13:59:26	30000000	0	35	Gmail	https://accounts.google.com/signin/v2/challenge/p...

Visit	URL ID	UTC	Trans.	From	Open	Title	URL
37	30	13:59:29	10000000	36	0	Inbox (12) - unsub.fscs@gmail.com -...	https://accounts.google.com/CheckCookie?continue=...
42	33	13:59:31	60000000	41	41	Inbox (12) - unsub.fscs@gmail.com -...	https://mail.google.com/mail/u/0/#inbox
43	34	14:04:12	10000000	0	42	cheaper than Rx supplements - order...	https://mail.google.com/mail/u/0/#inbox/WhctKKXXD...
44	35	14:04:35	40000000	43	43	cheaper than Rx supplements - order...	https://mail.google.com/mail/u/0/#inbox/WhctKKXXD...
45	36	14:04:46	40000000	44	44	cheaper than Rx supplements - order...	https://mail.google.com/mail/u/0/#inbox/WhctKKXXD...
46	34	14:04:52	60000000	45	45	cheaper than Rx supplements - order...	https://mail.google.com/mail/u/0/#inbox/WhctKKXXD...
47	33	14:05:59	30000000	0	46	Inbox (12) - unsub.fscs@gmail.com -...	https://mail.google.com/mail/u/0/#inbox
48	37	14:56:47	10000000	0	47	cheaper than Rx supplements -- txid...	https://mail.google.com/mail/u/0/#inbox/WhctKKXXD...
72	56	15:02:32	40000000	48	48	cheaper than Rx supplements - order...	https://mail.google.com/mail/u/0/#inbox/WhctKKXXD...
73	57	15:02:44	40000000	72	72	cheaper than Rx supplements - order...	https://mail.google.com/mail/u/0/#inbox/WhctKKXXD...
74	37	15:02:56	60000000	73	73	cheaper than Rx supplements -- txid...	https://mail.google.com/mail/u/0/#inbox/WhctKKXXD...

At 14:04:12 (visit 43), opened from the inbox, the profile opened the thread "cheaper than Rx supplements - order". Visits 44 and 45 are client redirects to addresses ending in ?compose=, and visit 46 returns to the thread at 14:04:52. This is consistent with a compose or reply pane being opened, but the history does not show that text was written or that a message was sent. The profile returned to the inbox at 14:05:59 (visit 47), and the next record is 50 minutes 48 seconds later.

5.6.2 The thread that carries the transaction identifier

At 14:56:47 (visit 48) the profile opened an inbox thread whose title is "cheaper than Rx supplements -- txid: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5". This is a different thread from the order thread, because the thread identifier in the address differs (it begins WhctKKXXDrfCqvtRjh in visit 48 and WhctKKXXDrfCqXBGC in visit 43). The record does not show who sent the message or whether the profile's account sent or received it.

5.6.3 Image hosting and the downloaded proof of payment

Visit 49 (14:56:52) is a Google link-wrapper address with source=gmail, opened from visit 48. It redirected to https://imgur.com/dTgrkP7 (visit 50), which stayed open for 18.9 seconds. The downloads table records one file, started six seconds later:

Field	Value (download id 1)
Saved as (target_path, current_path)	C:\Users\FSCS_User\Desktop\proof_of_payment.png
Started / ended (UTC)	2022-04-19 14:56:58 / 2022-04-19 14:57:10 (12 seconds)
Size and state	33844 of 33844 bytes; state 1 (complete); interrupt reason 0
MIME type	image/png
Source address (downloads_url_chains)	https://i.imgur.com/dTgrkP7.png
Page and referrer	https://imgur.com/dTgrkP7 ; https://imgur.com/
Server Last-Modified	Tue, 19 Apr 2022 14:52:06 GMT
ETag	"e4e316a7412fbc693e9268127fa8e1e6"
Opened flag / stored hash	0 (not opened from the Chrome downloads list) / empty (no hash stored)

Reading of the download. An image with the file name proof_of_payment.png was saved to the Windows desktop of the user profile FSCS_User, after the profile followed a link from the email thread. The Last-Modified time (14:52:06 GMT) comes from the web server and shows when the image was last changed there, about 4 minutes 52 seconds before the download started. The history contains no Imgur visit before 14:56:52, so any upload around 14:52 was not made through this profile, or the records are missing. The history cannot show what the image depicts, who made it, or whether it truly proves a payment, because no hash or content is stored.

5.6.4 Blockchain explorer and mempool

Visit 51 (14:58:59) is an omnibox search for blockchain explorer, leading to blockchain.com/explorer (visits 53 and 54, 14:59:01 and 14:59:02). At 14:59:06 the profile reached the search result and the Bitcoin transaction page for the identifier (visits 55 to 57). At 14:59:21 it opened a Bitcoin address page, 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt (visit 58). The record shows that these pages were loaded. It does not show what they displayed, whether the transaction was confirmed, or whose address the 38Rcs address is.

Visit	URL ID	UTC	Trans.	From	Open	Title	URL
51	40	14:58:59	32000005	0	0	blockchain explorer - Google Search	https://www.google.com/search?q=blockchain+explor...
53	41	14:59:01	30000000	52	0	Blockchain Explorer - Search the Bl...	https://www.blockchain.com/explorer

Visit	URL ID	UTC	Trans.	From	Open	Title	URL
54	41	14:59:02	30000000	0	53	Blockchain Explorer - Search the Bl...	https://www.blockchain.com/explorer
55	42	14:59:06	10000000	54	0	Transaction: 517b2156914944339a9613...	https://www.blockchain.com/search?search=517b2156...
56	43	14:59:06	A0000000	55	0	Transaction: 517b2156914944339a9613...	https://www.blockchain.com/btc/tx/517b21569149443...
57	43	14:59:06	30000000	54	56	Transaction: 517b2156914944339a9613...	https://www.blockchain.com/btc/tx/517b21569149443...
58	44	14:59:21	30000000	54	57	Address: 38RcsURWCDmCbYocmsZCnGz1Ft...	https://www.blockchain.com/btc/address/38RcsURWCD...

5.6.5 Exchange account and the second verification

Visit 59 (14:59:38) is an omnibox search for kraken. The profile opened the Kraken sign-in page at 14:59:47 (visit 61), then a page titled "Approve new device" at 15:00:21 (visit 63) that stayed open for 104.4 seconds. The funding, trade and instant-buy pages (visits 64 to 66, 15:01:11 to 15:01:21) were reached from visit 63. The title of visit 66 reads "41,543.80 USD - Kraken - Funding - Overview" and the title of the ledger page at 15:02:05 (visit 71) reads "41,563.70 USD - Kraken - History - Ledger". At 15:01:31 the profile opened a Kraken redirect link from the funding page (visit 67, opener visit 66) which led through visits 68 and 69 to https://mempool.space/tx/ for the same identifier (visit 70, 15:01:33).

Visit	URL ID	UTC	Trans.	From	Open	Title	URL
59	45	14:59:38	32000005	0	0	kraken - Google Search	https://www.google.com/search?q=kraken&rlz=1C1ONG...
61	46	14:59:47	30000000	60	0	Sign in to Kraken - Kraken | Buy, S...	https://www.kraken.com/en-us/sign-in
63	47	15:00:21	30000000	0	62	Approve new device - Kraken | Buy, ...	https://www.kraken.com/en-us/device-approval
64	48	15:01:11	30000000	63	0	(empty)	https://www.kraken.com/u/trade?i=true
65	49	15:01:12	30000000	63	64	Kraken | Buy, Sell and Margin Trade...	https://www.kraken.com/u/instant
66	50	15:01:21	30000000	63	65	41,543.80 USD - Kraken - Funding - ...	https://www.kraken.com/u/funding
67	51	15:01:31	10000000	0	66	Transaction: 517b2156914944339a9613...	https://www.kraken.com/redirect?url=https%3A%2F%2...
68	52	15:01:33	40000000	67	0	Transaction: 517b2156914944339a9613...	http://www.kraken.com/redirect?t=-_F22IhwssFD
69	53	15:01:33	80000000	68	0	Transaction: 517b2156914944339a9613...	https://www.kraken.com/redirect?t=-_F22IhwssFD
70	54	15:01:33	A0000000	69	0	Transaction: 517b2156914944339a9613...	https://mempool.space/tx/517b2156914944339a96137a...

Visit	URL ID	UTC	Trans.	From	Open	Title	URL
71	55	15:02:05	30000000	63	66	41,563.70 USD - Kraken - History - ...	https://www.kraken.com/u/history/ledger

The sign-in is recorded at 14:59:47, but the history does not show whether it succeeded or what was entered, and the 104.4 second duration only records how long the page stayed open. The two USD amounts are text from page titles. They do not establish who owns the account or why the figures differ by 19.90.

5.6.6 One transaction identifier across four services

The identifier 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 appears in nine visit records, in the email titles, the blockchain.com addresses and the mempool and Kraken redirect pages. The extracted 64-character value is identical in all nine.

Visit	UTC	Service	Found in	Interval from previous
48	14:56:47	Gmail thread (title)	title	-
55	14:59:06	blockchain.com search	URL	2 min 19 s
56	14:59:06	blockchain.com transaction page	URL	0 s
57	14:59:06	blockchain.com transaction page	URL	0 s
67	15:01:31	Kraken redirect link	URL	2 min 25 s
68	15:01:33	Kraken redirect page	title	2 s
69	15:01:33	Kraken redirect page	title	0 s
70	15:01:33	mempool.space transaction page	URL	0 s
74	15:02:56	Gmail thread (title)	title	1 min 23 s

The same identifier therefore links the email, two independent blockchain viewers and the exchange, and the session ends back on the email thread. The record does not show the amount, the sender or receiver of the transaction, or whether it relates to the listing.

Evidence references. E06 to E10; Figures S10, S11a, S11b, S12a to S12c, S13 and S14; queries Q11 to Q18.

Figure S10. Gmail and Google account visits (25 rows). [Evidence ID: E06]

SQL 1* ×

```
1  SELECT v.id, v.from_visit, v.opener_visit,
2   datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS utc,
3   printf('%08X', v.transition & 4294967295) AS tr_hex, substr(u.title,1,60) AS title, substr(u.url,1,60) AS url
4  FROM visits v JOIN urls u ON v.url=u.id
5  WHERE u.url LIKE '%mail.google.com%' OR u.url LIKE '%accounts.google.com%' ORDER BY v.visit_time;
```

	id	from_visit	opener_visit	utc	tr_hex	title	url
1	25	24	0	2022-04-19 13:58:40	80000000	Inbox (12) - unsub.fscs@gmail.com - …	https://mail.google.com/mail/
2	26	25	0	2022-04-19 13:58:40	80000000	Inbox (12) - unsub.fscs@gmail.com - …	https://mail.google.com/mail/u/0/
3	27	26	0	2022-04-19 13:58:40	80000000	Gmail: Free, Private & Secure Email …	https://accounts.google.com/…
4	28	27	0	2022-04-19 13:58:40	80000000	Gmail: Free, Private & Secure Email …	https://mail.google.com/intl/en/mail…
5	31	0	0	2022-04-19 13:58:43	10000000	Gmail	https://accounts.google.com/…
6	32	31	0	2022-04-19 13:58:43	80000000	Gmail	https://accounts.google.com/…
7	33	32	0	2022-04-19 13:58:43	A0000000	Gmail	https://accounts.google.com/…
8	34	0	33	2022-04-19 13:58:44	30000000	Gmail	https://accounts.google.com/…
9	35	0	34	2022-04-19 13:58:44	30000000	Gmail	https://accounts.google.com/signin/…
10	36	0	35	2022-04-19 13:59:26	30000000	Gmail	https://accounts.google.com/signin/…
11	37	36	0	2022-04-19 13:59:29	10000000	Inbox (12) - unsub.fscs@gmail.com - …	https://accounts.google.com/…
12	38	37	0	2022-04-19 13:59:29	80000000	Inbox (12) - unsub.fscs@gmail.com - …	https://mail.google.com/accounts/…
13	39	38	0	2022-04-19 13:59:29	80000000	Inbox (12) - unsub.fscs@gmail.com - …	https://accounts.youtube.com/…
14	40	39	0	2022-04-19 13:59:29	80000000	Inbox (12) - unsub.fscs@gmail.com - …	https://mail.google.com/mail/
15	41	40	0	2022-04-19 13:59:29	80000000	Inbox (12) - unsub.fscs@gmail.com - …	https://mail.google.com/mail/u/0/
16	42	41	41	2022-04-19 13:59:31	60000000	Inbox (12) - unsub.fscs@gmail.com - …	https://mail.google.com/mail/u/0/…

Capture rows 1 to 16 and 17 to 25.

Figure S11a. Visits containing the transaction identifier (9 rows). [Evidence ID: E08]

SQL 1* ×

```
1  SELECT v.id, datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS utc,
2   substr(u.title,1,70) AS title, substr(u.url,1,70) AS url
3  FROM visits v JOIN urls u ON v.url=u.id
4  WHERE u.url LIKE '%517b2156%' OR u.title LIKE '%517b2156%' ORDER BY v.visit_time;
```

	id	utc	title	url
1	48	2022-04-19 14:56:47	cheaper than Rx supplements -- txid:…	https://mail.google.com/mail/u/0/…
2	55	2022-04-19 14:59:06	Transaction: …	https://www.blockchain.com/search?…
3	56	2022-04-19 14:59:06	Transaction: …	https://www.blockchain.com/btc/tx/…
4	57	2022-04-19 14:59:06	Transaction: …	https://www.blockchain.com/btc/tx/…
5	67	2022-04-19 15:01:31	Transaction: …	https://www.kraken.com/redirect?…
6	68	2022-04-19 15:01:33	Transaction: …	http://www.kraken.com/redirect?t=-…
7	69	2022-04-19 15:01:33	Transaction: …	https://www.kraken.com/redirect?t=-…
8	70	2022-04-19 15:01:33	Transaction: …	https://mempool.space/tx/…
9	74	2022-04-19 15:02:56	cheaper than Rx supplements -- txid:…	https://mail.google.com/mail/u/0/…

Figure S11b. The extracted 64-character identifier on every row, with where it was found. [Evidence ID: E08]

SQL 1* ×

```
 CASE WHEN instr(u.url,'517b2156')>0 THEN 'url' ELSE 'title' END AS found_in
FROM visits v JOIN urls u ON v.url=u.id
WHERE u.url LIKE '%517b2156%' OR u.title LIKE '%517b2156%'
ORDER BY v.visit_time;
```

	id	utc	txid_found	found_in
1	48	2022-04-19 14:56:47	517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	title
2	55	2022-04-19 14:59:06	517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	url
3	56	2022-04-19 14:59:06	517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	url
4	57	2022-04-19 14:59:06	517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	url
5	67	2022-04-19 15:01:31	517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	url
6	68	2022-04-19 15:01:33	517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	title
7	69	2022-04-19 15:01:33	517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	title
8	70	2022-04-19 15:01:33	517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	url
9	74	2022-04-19 15:02:56	517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	title

To capture: Run Q13 in Appendix D. Widen the txid_found column so that all 64 characters are visible.

Figure S12a. Download record: file, times, size, state, tab address and referrer. [Evidence ID: E07]

SQL 1* ×

```
SELECT d.id, d.target_path, d.mime_type, d.state, d.received_bytes, d.total_bytes,
 datetime(d.start_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS start_utc,
 datetime(d.end_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS end_utc,
 d.tab_url, d.referrer, d.last_modified
FROM downloads d;
```

	id	target_path	mime_type	state	received_bytes	otal_byte	start_utc	end_utc	tab_url	referrer	
1	1	C:…	image/png	1	33844	33844	2022-04-19 14:56:58	2022-04-19 14:57:10	https://imgur.com/dTgrkP7	https://imgur.com/	Tue, 19

To capture: Run Q14 in Appendix D.

Figure S12b. Download source address (downloads_url_chains). [Evidence ID: E07]

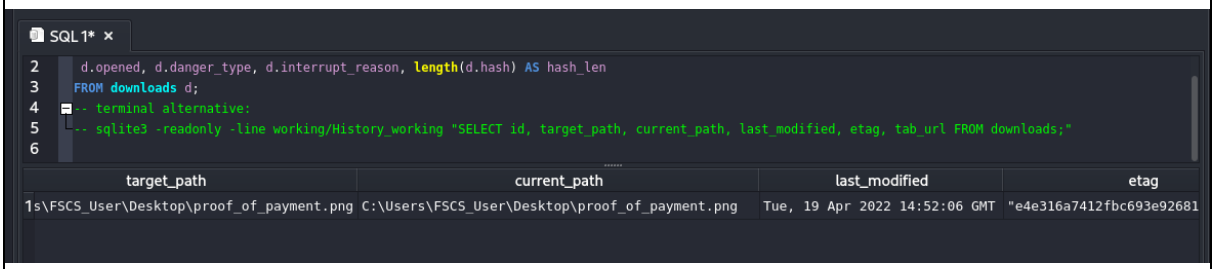
SQL 1* ×

```
SELECT id, chain_index, url FROM downloads_url_chains;
```

	id	chain_index	url
1	1	0	https://i.imgur.com/dTgrkP7.png

To capture: Run Q15 in Appendix D.

Figure S12c. Full saved path, Last-Modified, ETag and flags of the download. [Evidence ID: E07]

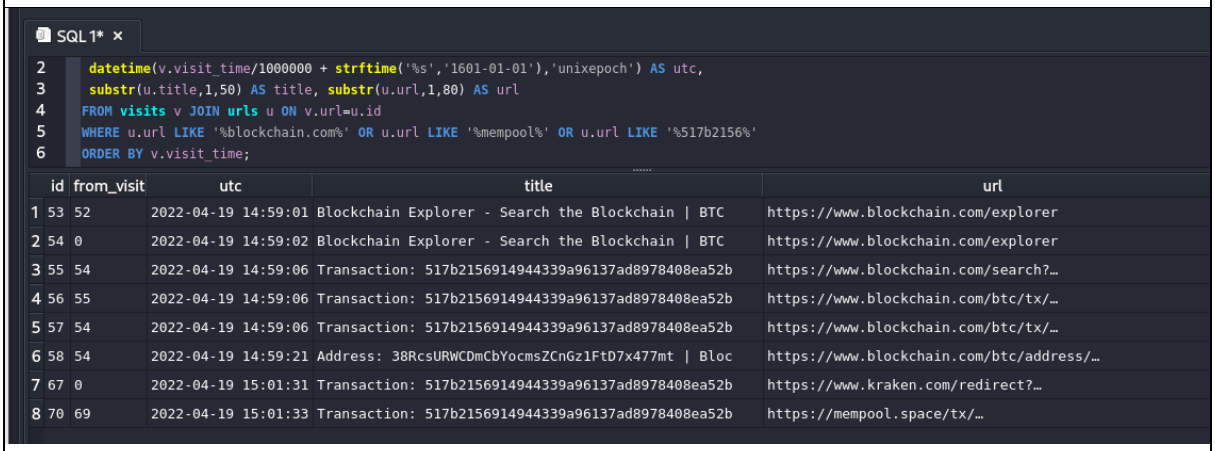


```
SQL 1* ×

  d.opened, d.danger_type, d.interrupt_reason, length(d.hash) AS hash_len
FROM downloads d;
-- terminal alternative:
-- sqlite3 -readonly -line working/History_working "SELECT id, target_path, current_path, last_modified, etag, tab_url FROM downloads;"
```

	target_path	current_path	last_modified	etag
1	s\FSCS_User\Desktop\proof_of_payment.png	C:\Users\FSCS_User\Desktop\proof_of_payment.png	Tue, 19 Apr 2022 14:52:06 GMT	"e4e316a7412fbc693e92681

Figure S13. Blockchain explorer and mempool visits (8 rows). [Evidence ID: E09]

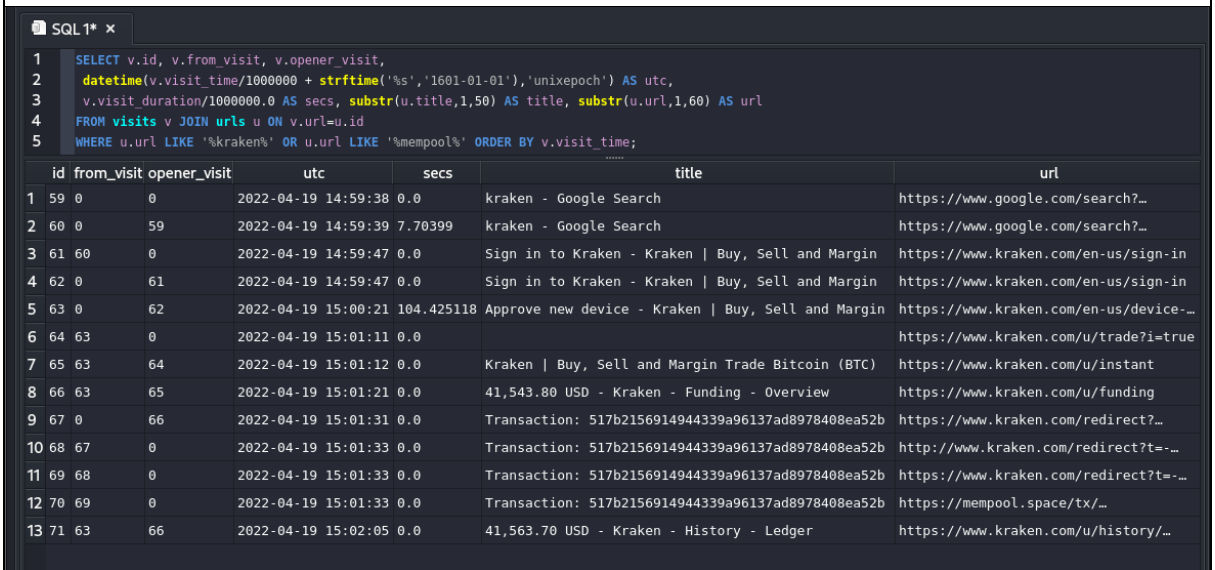


```
SQL 1* ×

  datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS utc,
  substr(u.title,1,50) AS title, substr(u.url,1,80) AS url
FROM visits v JOIN urls u ON v.url=u.id
WHERE u.url LIKE '%blockchain.com%' OR u.url LIKE '%mempool%' OR u.url LIKE '%517b2156%'
ORDER BY v.visit_time;
```

	id	from_visit	utc	title	url
1	53	52	2022-04-19 14:59:01	Blockchain Explorer - Search the Blockchain | BTC	https://www.blockchain.com/explorer
2	54	0	2022-04-19 14:59:02	Blockchain Explorer - Search the Blockchain | BTC	https://www.blockchain.com/explorer
3	55	54	2022-04-19 14:59:06	Transaction: 517b2156914944339a96137ad8978408ea52b	https://www.blockchain.com/search?…
4	56	55	2022-04-19 14:59:06	Transaction: 517b2156914944339a96137ad8978408ea52b	https://www.blockchain.com/btc/tx/…
5	57	54	2022-04-19 14:59:06	Transaction: 517b2156914944339a96137ad8978408ea52b	https://www.blockchain.com/btc/tx/…
6	58	54	2022-04-19 14:59:21	Address: 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt | Bloc	https://www.blockchain.com/btc/address/…
7	67	0	2022-04-19 15:01:31	Transaction: 517b2156914944339a96137ad8978408ea52b	https://www.kraken.com/redirect?…
8	70	69	2022-04-19 15:01:33	Transaction: 517b2156914944339a96137ad8978408ea52b	https://mempool.space/tx/…

Figure S14. Kraken and mempool visits with durations (13 rows). [Evidence ID: E10]



```
SQL 1* ×

SELECT v.id, v.from_visit, v.opener_visit,
  datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS utc,
  v.visit_duration/1000000.0 AS secs, substr(u.title,1,50) AS title, substr(u.url,1,60) AS url
FROM visits v JOIN urls u ON v.url=u.id
WHERE u.url LIKE '%kraken%' OR u.url LIKE '%mempool%' ORDER BY v.visit_time;
```

	id	from_visit	opener_visit	utc	secs	title	url
1	59	0	0	2022-04-19 14:59:38	0.0	kraken - Google Search	https://www.google.com/search?…
2	60	0	59	2022-04-19 14:59:39	7.70399	kraken - Google Search	https://www.google.com/search?…
3	61	60	0	2022-04-19 14:59:47	0.0	Sign in to Kraken - Kraken | Buy, Sell and Margin	https://www.kraken.com/en-us/sign-in
4	62	0	61	2022-04-19 14:59:47	0.0	Sign in to Kraken - Kraken | Buy, Sell and Margin	https://www.kraken.com/en-us/sign-in
5	63	0	62	2022-04-19 15:00:21	104.425118	Approve new device - Kraken | Buy, Sell and Margin	https://www.kraken.com/en-us/device-…
6	64	63	0	2022-04-19 15:01:11	0.0		https://www.kraken.com/u/trade?i=true
7	65	63	64	2022-04-19 15:01:12	0.0	Kraken | Buy, Sell and Margin Trade Bitcoin (BTC)	https://www.kraken.com/u/instant
8	66	63	65	2022-04-19 15:01:21	0.0	41,543.80 USD - Kraken - Funding - Overview	https://www.kraken.com/u/funding
9	67	0	66	2022-04-19 15:01:31	0.0	Transaction: 517b2156914944339a96137ad8978408ea52b	https://www.kraken.com/redirect?…
10	68	67	0	2022-04-19 15:01:33	0.0	Transaction: 517b2156914944339a96137ad8978408ea52b	http://www.kraken.com/redirect?t=-…
11	69	68	0	2022-04-19 15:01:33	0.0	Transaction: 517b2156914944339a96137ad8978408ea52b	https://www.kraken.com/redirect?t=-…
12	70	69	0	2022-04-19 15:01:33	0.0	Transaction: 517b2156914944339a96137ad8978408ea52b	https://mempool.space/tx/…
13	71	63	66	2022-04-19 15:02:05	0.0	41,563.70 USD - Kraken - History - Ledger	https://www.kraken.com/u/history/…

5.7 Question 7: What is the reconstructed timeline?

The reconstructed timeline separates the artefact, the observed fact and the interpretation, and is set out in Section 6. The complete record of all 74 visits is in Appendix B.

5.8 Question 8: What conclusion can you defend?

The defensible conclusion, its confidence levels and its limitations are set out in Section 7. In short: the history shows that one Chrome profile performed a sequence of actions consistent with a posting, communication, payment-information exchange, transaction verification and retention of a proof-of-payment image. It does not prove the content of any page, the completion of any transaction, that the goods were controlled drugs, or who used the device.

6. Timeline

All times are UTC on 19 April 2022. Lagos time (UTC+1) is one hour later. The column "Observed fact" reports only what a record states. The column "Interpretation and limit" is the examiner's reading and its limit.

Stage	UTC timestamp	Visit or URL ID	Artefact	Observed fact	Interpretation and limit
1. Search and login	13:54:18 to 13:54:39	Visits 1 to 8; URLs 1 to 6	Google search, Craigslist account login page	Omnibox search craigslist (v1); login page open 17.5 s (v7); form submit (v8)	Consistent with logging in to an account. Success of the login and the account holder are not shown.
2. Posting	13:55:05 to 13:58:22	Visits 11 to 19; URLs 7 to 14	Craigslist posting wizard	Six form submissions (v14 to v19); create page open 177.7 s (v15); confirmation at v19	Consistent with creating an advert. Text and images are not stored. Visit 12 is a redirect, not the publication.
3. Publication	13:58:27 to 13:58:31	Visits 20, 21; URLs 15, 16	Manage posting 7473121658; live listing page	Listing titled cheaper than Rx supplements, health and beauty, by owner, opened from v20	The advert was published. Its wording is unknown, and the title refers to supplements, not to controlled drugs.
4. Email access	13:58:38 to 13:59:31	Visits 22 to 42; URLs 17 to 33	Omnibox search gmail; sign-in; inbox of unsub.fscs@gmail.com	Redirect chains; password-challenge page at 13:59:26 (v36); inbox at 13:59:31 (v42)	The profile reached a Gmail account. Account ownership is not proven. Titles are latest-per-URL.
5. Buyer contact	14:04:12 to 14:05:59	Visits 43 to 47; URLs 34, 35, 36, 33	Thread "order"; compose addresses	Thread opened from inbox (v43); compose redirects (v44, v45); return (v46)	A reply may have been opened. No proof that anything was written or sent. No records follow for 50 min 48 s.
6. Payment information	14:56:47	Visit 48; URL 37	A different Gmail thread whose subject contains the transaction identifier	Title includes txid: 517b2156...	A message carrying a payment reference is present. Sender and direction are unknown.

Stage	UTC timestamp	Visit or URL ID	Artefact	Observed fact	Interpretation and limit
7. Proof of payment	14:56:52 to 14:57:10	Visits 49, 50; URLs 38, 39; download 1	Imgur page; proof_of_payment.png	Link opened from the thread; 33,844 byte PNG saved to the Desktop of FSCS_User; server Last-Modified 14:52:06 GMT	An image was retained locally. Its content, author and upload route are unknown. No hash is stored.
8. Public chain check	14:58:59 to 14:59:21	Visits 51 to 58; URLs 40 to 44	Omnibox search blockchain explorer; transaction page; address page	Same identifier searched at 14:59:06; address 38RcsURW...x477mt opened at 14:59:21	A lookup was made. The result is not stored. Address ownership is unknown.
9. Exchange account	14:59:38 to 15:02:05	Visits 59 to 66, 71; URLs 45 to 50, 55	Omnibox search kraken; sign-in; device approval; funding; ledger	Device-approval page open 104.4 s; titles show 41,543.80 USD and 41,563.70 USD	An exchange account area was reached. Ownership and the cause of the figures are not shown.
10. Second verification	15:01:31 to 15:01:33	Visits 67 to 70; URLs 51 to 54	Kraken redirect to mempool.space transaction page	Same identifier; opened from the funding page (v66)	A second viewer was used. The outcome is not stored.
11. Follow-up	15:02:32 to 15:02:56	Visits 72 to 74; URLs 56, 57, 37	Compose addresses chained from v48; txid thread	Client redirects to ?compose= then the thread; session ends at 15:02:56	A reply may have been opened. Not proof of sending.

7. Attribution Assessment

7.1 Confidence

Statement	Confidence	Basis
One Chrome profile performed the whole recorded sequence	Moderate to high	A single History file; continuous 68 min 38 s; visits linked by from_visit and opener_visit; one identifier across the sequence. A History file belongs to one profile.
A Craigslist listing was created and published	High	Posting wizard (v11 to v19), management page ID 7473121658 (v20) and listing page (v21).
The listing advertised controlled drugs	Low	The title and category refer to supplements. The drug element comes from the case hypothesis.
Email communication about an order and a payment reference took place	Moderate (existence) / Low (content)	Thread titles (v43, v48) and compose addresses. No message text or sender is stored.
A proof-of-payment image was downloaded	High (file and name) / Low (content)	Download record 1, complete, 33,844 bytes, no stored hash.
A cryptocurrency payment occurred	Low	The identifier was looked up; no amount, parties or confirmation are stored.

Statement	Confidence	Basis
The identity of the person using the device	Low	Account labels and a Windows user name are not proof of a person (Section 7.2).
Overall: the record is consistent with the case hypothesis	Moderate	The sequence fits the hypothesis closely but rests on titles and URLs only.

7.2 Limitations and alternative explanations

- **Titles and URLs, not content.** The history shows addresses visited and titles. It does not store page text, email bodies, listing wording or image content.
- **Latest title per URL.** The urls table keeps one title for each address, and it is overwritten on later visits. Titles shown for earlier visits may not be what was displayed at the time (for example visits 25 and 26).
- **Browser profile is not a person.** The History file identifies a Chrome profile. The Windows user name FSCS_User, the Gmail label and the Kraken session do not show who was at the keyboard. A shared computer, a shared profile, remote access or a borrowed session are all possible.
- **Redirects and automatic navigation.** Chrome records redirect steps and automatic page loads as visits. Visits 24 to 30, 37 to 42 and 67 to 70 are chains and not separate user actions.
- **Compose addresses are not messages.** A ?compose= address shows a compose or reply pane opened, not that a message was written or sent.
- **Missing content.** No records exist between 14:05:59 and 14:56:47. History can be cleared, edited, or recorded on another profile or device, and a session that moved to another browser would not appear.
- **Cached content not examined.** The browser cache, cookies, the cached pages and the saved image itself would carry more detail than the History file. They were not part of the supplied database.
- **The transaction.** Visiting a page that carries a transaction identifier does not show that the transaction exists, completed, or relates to the listing. The identifier appears in pages the profile loaded, not in a record of value transferred.
- **Server Last-Modified.** The 14:52:06 GMT value comes from the web server and may not reflect the true creation time of the image.
- **Account figures.** The USD amounts are in page titles and may not match the account balance or any transfer.
- **Device clock.** All times depend on the clock of the device that created the History file. Chrome stores them in UTC, but a wrong clock would shift every record.

Alternative explanations. (1) The activity could relate to the legitimate sale of supplements, because nothing in the titles states otherwise. (2) The profile could belong to someone who

was testing, researching or training on these services. (3) The proof-of-payment thread could belong to a purchase unrelated to the listing. (4) The profile could have been used by more than one person. The record cannot exclude any of these.

7.3 Recommended follow-up

- Obtain, through proper legal process, the content and account records for Craigslist posting 7473121658, the Gmail account that appears in the page titles, the Imgur image dTgrkP7 and the Kraken account.
- Examine the source device, subject to authority: the saved file proof_of_payment.png (hash and content), Chrome cache and cookies, other browser profiles, and the Windows user accounts.
- Analyse the transaction on the public chain using lawful tools to establish the amount, the addresses and the time, and compare with the listing.

8. Appendices

Appendix A: Screenshot Index

Screenshots are inserted in the figure boxes in Sections 3 to 5. Each capture should show the SQL that produced the result (or the command run) and the result. Figure S15 is not used.

Figure	Content	Evidence ID	Section
S01	Hash and size of the original file	E01	3
S02	Both hashes after the working copy was created	E01	3
S03	Database Structure view, 17 tables	E02	4.1
S04, S04b	Field definitions; record counts	E02	4.1
S05	Timestamp conversion, first and last visit	E03	5.3
S06a to S06e	Visit timeline, 74 rows	E03	5.3
S07	Transition decode	E04	5.4
S08a to S08d	Distribution; from_visit chain; opener visits; search terms	E04	5.4
S09	Craigslist visits (21 rows)	E05	5.5
S10	Gmail and Google account visits (25 rows)	E06	5.6
S11a, S11b	Transaction identifier across visits; extracted value	E08	5.6
S12a to S12c	Download record, source address and full path	E07	5.6
S13	Blockchain explorer and mempool (8 rows)	E09	5.6
S14	Kraken and mempool (13 rows)	E10	5.6

Appendix B: Complete Visit Timeline (74 visits, UTC, 19 April 2022)

Columns: visit id; URL id; UTC time; transition value in hexadecimal; from_visit; opener_visit; page title and address, shortened. All values were produced by query Q05 and Q06 on the working copy.

Visit	URL	UTC	Trans.	From	Open	Title	URL (shortened)
1	1	13:54:18	30000005	0	0	craigslist - Google Search	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_...
2	1	13:54:19	30000000	0	1	craigslist - Google Search	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_...
3	2	13:54:19	10000000	2	0	craigslist: baltimore jobs, apartments, for...	https://www.craigslist.org/
4	3	13:54:19	80000000	3	0	craigslist: baltimore jobs, apartments, for...	https://geo.craigslist.org/
5	4	13:54:19	A0000000	4	0	craigslist: baltimore jobs, apartments, for...	https://baltimore.craigslist.org/
6	5	13:54:22	10000000	5	0	craigslist account	https://accounts.craigslist.org/login/home
7	6	13:54:22	A0000000	6	0	craigslist - account log in	https://accounts.craigslist.org/login?rp=%2Flogin%2Fhom...
8	5	13:54:39	30000007	7	0	craigslist account	https://accounts.craigslist.org/login/home
9	2	13:55:03	10000000	8	0	craigslist: baltimore jobs, apartments, for...	https://www.craigslist.org/
10	4	13:55:03	A0000000	9	0	craigslist: baltimore jobs, apartments, for...	https://baltimore.craigslist.org/
11	7	13:55:05	10000000	10	0	baltimore | choose type	https://post.craigslist.org/c/bal
12	8	13:55:05	80000000	11	0	baltimore | posting confirmation	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5
13	9	13:55:05	A0000000	12	0	baltimore | choose type	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQ...
14	10	13:55:10	30000007	13	0	baltimore | choose category	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQ...
15	11	13:55:15	30000007	14	0	baltimore | create posting	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQ...
16	12	13:58:13	30000007	15	0	baltimore | add map	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQ...
17	13	13:58:17	30000007	16	0	baltimore | create posting	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQ...
18	14	13:58:19	30000007	17	0	baltimore | create posting	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQ...
19	8	13:58:22	30000007	18	0	baltimore | posting confirmation	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5
20	15	13:58:27	30000000	19	0	baltimore | manage posting	https://post.craigslist.org/manage/7473121658
21	16	13:58:31	30000000	0	20	cheaper than Rx supplements - health and be...	https://baltimore.craigslist.org/hab/d/baltimore-cheape...
22	17	13:58:38	32000005	0	0	gmail - Google Search	https://www.google.com/search?q=gmail&rlz=1C1ONGR_enUS9...
23	17	13:58:39	30000000	0	22	gmail - Google Search	https://www.google.com/search?q=gmail&rlz=1C1ONGR_enUS9...
24	18	13:58:40	10000000	23	0	Gmail: Free, Private & Secure Email | Googl...	https://www.google.com/gmail/
25	19	13:58:40	80000000	24	0	Inbox (12) - unsub.fscs@gmail.com - Gmail	https://mail.google.com/mail/
26	20	13:58:40	80000000	25	0	Inbox (12) - unsub.fscs@gmail.com - Gmail	https://mail.google.com/mail/u/0/
27	21	13:58:40	80000000	26	0	Gmail: Free, Private & Secure Email | Googl...	https://accounts.google.com/ServiceLogin?service=mail&p...
28	22	13:58:40	80000000	27	0	Gmail: Free, Private & Secure Email | Googl...	https://mail.google.com/intl/en/mail/help/about.html

Visit	URL	UTC	Trans.	From	Open	Title	URL (shortened)
29	23	13:58:40	80000000	28	0	Gmail: Free, Private & Secure Email | Googl...	https://www.google.com/intl/en/mail/help/about.html
30	24	13:58:40	A0000000	29	0	Gmail: Free, Private & Secure Email | Googl...	https://www.google.com/gmail/about/
31	25	13:58:43	10000000	0	0	Gmail	https://accounts.google.com/AccountChooser/signinchoose...
32	26	13:58:43	80000000	31	0	Gmail	https://accounts.google.com/AccountChooser?service=mail...
33	27	13:58:43	A0000000	32	0	Gmail	https://accounts.google.com/ServiceLogin?continue=https...
34	27	13:58:44	30000000	0	33	Gmail	https://accounts.google.com/ServiceLogin?continue=https...
35	28	13:58:44	30000000	0	34	Gmail	https://accounts.google.com/signin/v2/identifier?contin...
36	29	13:59:26	30000000	0	35	Gmail	https://accounts.google.com/signin/v2/challenge/pwd?con...
37	30	13:59:29	10000000	36	0	Inbox (12) - unsub.fscs@gmail.com - Gmail	https://accounts.google.com/CheckCookie?continue=https%...
38	31	13:59:29	80000000	37	0	Inbox (12) - unsub.fscs@gmail.com - Gmail	https://mail.google.com/accounts/SetOSID?authuser=0&con...
39	32	13:59:29	80000000	38	0	Inbox (12) - unsub.fscs@gmail.com - Gmail	https://accounts.youtube.com/accounts/SetSID?ssdc=1&sid...
40	19	13:59:29	80000000	39	0	Inbox (12) - unsub.fscs@gmail.com - Gmail	https://mail.google.com/mail/
41	20	13:59:29	80000000	40	0	Inbox (12) - unsub.fscs@gmail.com - Gmail	https://mail.google.com/mail/u/0/
42	33	13:59:31	60000000	41	41	Inbox (12) - unsub.fscs@gmail.com - Gmail	https://mail.google.com/mail/u/0/#inbox
43	34	14:04:12	10000000	0	42	cheaper than Rx supplements - order - unsub...	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXB...
44	35	14:04:35	40000000	43	43	cheaper than Rx supplements - order - unsub...	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXB...
45	36	14:04:46	40000000	44	44	cheaper than Rx supplements - order - unsub...	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXB...
46	34	14:04:52	60000000	45	45	cheaper than Rx supplements - order - unsub...	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXB...
47	33	14:05:59	30000000	0	46	Inbox (12) - unsub.fscs@gmail.com - Gmail	https://mail.google.com/mail/u/0/#inbox
48	37	14:56:47	10000000	0	47	cheaper than Rx supplements -- txid: 517b21...	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvt...
49	38	14:56:52	10000000	0	48	Imgur: The magic of the Internet	https://www.google.com/url?q=https://imgur.com/dTgrkP7&...
50	39	14:56:52	A0000000	49	0	Imgur: The magic of the Internet	https://imgur.com/dTgrkP7
51	40	14:58:59	32000005	0	0	blockchain explorer - Google Search	https://www.google.com/search?q=blockchain+explorer&rlz...
52	40	14:59:00	30000000	0	51	blockchain explorer - Google Search	https://www.google.com/search?q=blockchain+explorer&rlz...
53	41	14:59:01	30000000	52	0	Blockchain Explorer - Search the Blockchain...	https://www.blockchain.com/explorer
54	41	14:59:02	30000000	0	53	Blockchain Explorer - Search the Blockchain...	https://www.blockchain.com/explorer
55	42	14:59:06	10000000	54	0	Transaction: 517b2156914944339a96137ad89784...	https://www.blockchain.com/search?search=517b2156914944...
56	43	14:59:06	A0000000	55	0	Transaction: 517b2156914944339a96137ad89784...	https://www.blockchain.com/btc/tx/517b2156914944339a961...
57	43	14:59:06	30000000	54	56	Transaction: 517b2156914944339a96137ad89784...	https://www.blockchain.com/btc/tx/517b2156914944339a961...

Visit	URL	UTC	Trans.	From	Open	Title	URL (shortened)
58	44	14:59:21	30000000	54	57	Address: 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt...	https://www.blockchain.com/btc/address/38RcsURWCDmCbYoc...
59	45	14:59:38	32000005	0	0	kraken - Google Search	https://www.google.com/search?q=kraken&rlz=1C1ONGR_enUS...
60	45	14:59:39	30000000	0	59	kraken - Google Search	https://www.google.com/search?q=kraken&rlz=1C1ONGR_enUS...
61	46	14:59:47	30000000	60	0	Sign in to Kraken - Kraken | Buy, Sell and ...	https://www.kraken.com/en-us/sign-in
62	46	14:59:47	30000000	0	61	Sign in to Kraken - Kraken | Buy, Sell and ...	https://www.kraken.com/en-us/sign-in
63	47	15:00:21	30000000	0	62	Approve new device - Kraken | Buy, Sell and...	https://www.kraken.com/en-us/device-approval
64	48	15:01:11	30000000	63	0	(empty)	https://www.kraken.com/u/trade?i=true
65	49	15:01:12	30000000	63	64	Kraken | Buy, Sell and Margin Trade Bitcoin...	https://www.kraken.com/u/instant
66	50	15:01:21	30000000	63	65	41,543.80 USD - Kraken - Funding - Overview	https://www.kraken.com/u/funding
67	51	15:01:31	10000000	0	66	Transaction: 517b2156914944339a96137ad89784...	https://www.kraken.com/redirect?url=https%3A%2F%2Fmempo...
68	52	15:01:33	40000000	67	0	Transaction: 517b2156914944339a96137ad89784...	http://www.kraken.com/redirect?t=-_F22IhwssFD
69	53	15:01:33	80000000	68	0	Transaction: 517b2156914944339a96137ad89784...	https://www.kraken.com/redirect?t=-_F22IhwssFD
70	54	15:01:33	A0000000	69	0	Transaction: 517b2156914944339a96137ad89784...	https://mempool.space/tx/517b2156914944339a96137ad89784...
71	55	15:02:05	30000000	63	66	41,563.70 USD - Kraken - History - Ledger	https://www.kraken.com/u/history/ledger
72	56	15:02:32	40000000	48	48	cheaper than Rx supplements - order - unsub...	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvt...
73	57	15:02:44	40000000	72	72	cheaper than Rx supplements - order - unsub...	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvt...
74	37	15:02:56	60000000	73	73	cheaper than Rx supplements -- txid: 517b21...	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvt...

Appendix C: Evidence Log

Evidence ID	Source record	Finding	Why it matters	SQL or screenshot reference
E01	File History; working copy	196,608 bytes; SHA-256 b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19; original read-only; copy identical	Identifies the exact evidence analysed and shows its integrity	S01, S02
E02	Tables urls, visits, downloads, downloads_url_chains	17 tables; 57 URLs, 74 visits, 1 download, 1 chain	Shows which tables and fields carry the evidence	S03, S04, S04b; Q01 to Q03
E03	visits 1 to 74 (visit_time)	All activity on 19 April 2022, 13:54:18 to 15:02:56 UTC (1 h 8 min 38 s); Lagos time is one hour later	Fixes the period and order of the activity	S05, S06a to S06e; Q04, Q05; Appendix B
E04	visits.transition, from_visit, opener_visit; keyword_search_terms	63 links, 4 omnibox searches, 7 form submits; 12 redirect and compose chains; listing, Imgur page and mempool page opened from earlier visits	Shows how each page was reached and separates user actions from redirects	S07, S08a to S08d; Q06 to Q09, Q19
E05	visits 1 to 21; urls 1 to 16	Craigslist login, posting wizard (v14 to v19), management page 7473121658 (v20) and listing (v21) at 13:58:31	Supports the posting of an advert	S09; Q10
E06	visits 22 to 47; 72 to 74	Gmail sign-in and inbox for unsub.fscs@gmail.com; thread "order" (v43) with compose redirects	Supports communication with a counterpart	S10; Q11
E07	downloads id 1; downloads_url_chains id 1; visits 49, 50	proof_of_payment.png, 33,844 bytes, complete, saved to Desktop 14:56:58 to 14:57:10 from i.imgur.com/dTgrkP7.png; Last-Modified 14:52:06 GMT	Shows retention of a proof-of-payment file	S12a, S12b, S12c; Q14 to Q16
E08	visits 48, 55 to 57, 67 to 70, 74	One transaction identifier (517b2156...afdc5) in nine records across Gmail, blockchain.com, Kraken redirect and mempool.space	Links the email, the explorer and the exchange	S11a, S11b; Q12, Q13
E09	visits 51, 53 to 58, 67, 70	Omnibox search blockchain explorer; transaction page (14:59:06); address page 38RcsURW...x477mt (14:59:21)	Supports verification of a cryptocurrency transaction	S13; Q17
E10	visits 59 to 71	Omnibox search kraken; sign-in; device approval 104.4 s; funding and ledger pages; redirect to mempool.space (15:01:33)	Supports exchange activity and a second verification	S14; Q18

Appendix D: SQL Reference

All queries were run on working/History_working with the database open read-only. In sqlite3 each query is run with: sqlite3 -readonly -header -column working/History_working "<query>".

Q01 (S03): list tables

```
SELECT name FROM sqlite_master WHERE type='table' ORDER BY name;
```

Q02 (S04): table definitions

```
PRAGMA table_info(urls);
PRAGMA table_info(visits);
PRAGMA table_info(downloads);
PRAGMA table_info(downloads_url_chains);
```

Q03 (S04b): record counts

```
SELECT 'urls' AS tbl,COUNT(*) AS n FROM urls
UNION ALL SELECT 'visits',COUNT(*) FROM visits
UNION ALL SELECT 'downloads',COUNT(*) FROM downloads
UNION ALL SELECT 'downloads_url_chains',COUNT(*) FROM downloads_url_chains;
```

Q04 (S05): timestamp conversion

```
SELECT id, visit_time AS raw_webkit,
 datetime(visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS utc,
 datetime(visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch','+1 hour') AS lagos_wat
FROM visits WHERE id IN (1,74);
```

Q05 (S06): visit timeline

```
SELECT v.id AS visit_id, u.id AS url_id,
 datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS utc,
 v.from_visit, v.opener_visit, v.transition,
 substr(u.title,1,40) AS title, substr(u.url,1,55) AS url
FROM visits v JOIN urls u ON v.url=u.id
ORDER BY v.visit_time;
```

Q06 (S07): transition decode

```
SELECT v.id, v.from_visit, v.opener_visit,
 printf('%08X', v.transition & 4294967295) AS tr_hex,
 v.transition & 255 AS base,
 CASE v.transition & 255 WHEN 0 THEN 'LINK' WHEN 5 THEN 'OMNIBOX'
  WHEN 7 THEN 'FORM_SUBMIT' ELSE 'other' END AS base_name,
 (v.transition & 268435456)<>0 AS chain_start,
 (v.transition & 536870912)<>0 AS chain_end,
 (v.transition & 1073741824)<>0 AS client_redir,
 (v.transition & 2147483648)<>0 AS server_redir,
 (v.transition & 33554432)<>0 AS addr_bar
FROM visits v ORDER BY v.visit_time;
```

Q07 (S08a): base type distribution

```
SELECT transition & 255 AS base, COUNT(*) AS n FROM visits GROUP BY base;
```

Q08 (S08b): from_visit chain

```
SELECT v.id, v.from_visit, substr(pu.url,1,45) AS came_from, substr(u.url,1,45) AS landed_on,
 printf('%08X', v.transition & 4294967295) AS tr_hex
FROM visits v JOIN urls u ON v.url=u.id
LEFT JOIN visits pv ON v.from_visit=pv.id LEFT JOIN urls pu ON pv.url=pu.id
WHERE v.from_visit<>0 ORDER BY v.visit_time;
```

Q09 (S08c): opener visits

```
SELECT v.id, v.opener_visit, substr(u.url,1,50) AS landed_on
FROM visits v JOIN urls u ON v.url=u.id
WHERE v.opener_visit<>0 ORDER BY v.visit_time;
```

Q10 (S09): Craigslist

```
SELECT v.id, u.id AS url_id,
 datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS utc,
 v.transition & 255 AS base, u.title, substr(u.url,1,70) AS url
FROM visits v JOIN urls u ON v.url=u.id
WHERE u.url LIKE '%craigslist%' ORDER BY v.visit_time;
```

Q11 (S10): Gmail and Google account

```
SELECT v.id, v.from_visit, v.opener_visit,
 datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS utc,
 printf('%08X', v.transition & 4294967295) AS tr_hex,
 substr(u.title,1,60) AS title, substr(u.url,1,60) AS url
FROM visits v JOIN urls u ON v.url=u.id
WHERE u.url LIKE '%mail.google.com%' OR u.url LIKE '%accounts.google.com%'
ORDER BY v.visit_time;
```

Q12 (S11a): transaction identifier

```
SELECT v.id,
 datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS utc,
 substr(u.title,1,70) AS title, substr(u.url,1,70) AS url
FROM visits v JOIN urls u ON v.url=u.id
WHERE u.url LIKE '%517b2156%' OR u.title LIKE '%517b2156%'
ORDER BY v.visit_time;
```

Q13 (S11b): extracted identifier

```
SELECT v.id,
 datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS utc,
 CASE WHEN instr(u.url,'517b2156')>0 THEN substr(u.url,instr(u.url,'517b2156'),64)
     ELSE substr(u.title,instr(u.title,'517b2156'),64) END AS txid_found,
 CASE WHEN instr(u.url,'517b2156')>0 THEN 'url' ELSE 'title' END AS found_in
FROM visits v JOIN urls u ON v.url=u.id
WHERE u.url LIKE '%517b2156%' OR u.title LIKE '%517b2156%'
ORDER BY v.visit_time;
```

Q14 (S12a): download record

```
SELECT d.id, d.target_path, d.mime_type, d.state, d.received_bytes, d.total_bytes,
 datetime(d.start_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS start_utc,
 datetime(d.end_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS end_utc,
 d.tab_url, d.referrer, d.last_modified
FROM downloads d;
```

Q15 (S12b): download source

```
SELECT id, chain_index, url FROM downloads_url_chains;
```

Q16 (S12c): full path and flags

```
SELECT d.target_path, d.current_path, d.last_modified, d.etag,
 d.opened, d.danger_type, d.interrupt_reason, length(d.hash) AS hash_len
FROM downloads d;
-- terminal alternative:
-- sqlite3 -readonly -line working/History_working "SELECT id, target_path, current_path, last_modified,
etag, tab_url FROM downloads;"
```

Q17 (S13): blockchain and mempool

```
SELECT v.id, v.from_visit,
 datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS utc,
 substr(u.title,1,50) AS title, substr(u.url,1,80) AS url
FROM visits v JOIN urls u ON v.url=u.id
WHERE u.url LIKE '%blockchain.com%' OR u.url LIKE '%mempool%' OR u.url LIKE '%517b2156%'
ORDER BY v.visit_time;
```

Q18 (S14): Kraken and mempool

```
SELECT v.id, v.from_visit, v.opener_visit,
 datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS utc,
 v.visit_duration/1000000.0 AS secs, substr(u.title,1,50) AS title, substr(u.url,1,60) AS url
FROM visits v JOIN urls u ON v.url=u.id
WHERE u.url LIKE '%kraken%' OR u.url LIKE '%mempool%' ORDER BY v.visit_time;
```

Q19 (S08d): search terms

```
SELECT keyword_id, url_id, term, normalized_term FROM keyword_search_terms;
```

Appendix E: Redaction Statement

Long query strings and address fragments that act as session tokens (for example in Google sign-in and Gmail addresses) were shortened and are not reproduced in this report. The Gmail account label, the Windows user name, the posting identifier and the transaction identifier were kept because they are needed to verify the findings. Unrelated personal data, if it appears in a screenshot, should be blacked out before the image is inserted.